ClassGuard

App Guide & Project Status

A plain-language guide to the ClassGuard camera management desktop app: what it does, how it works, what is finished, and what still needs to be built.

Hikvision · Dahua · UNV · CP Plus · ONVIF · Any RTSP camera · NVR / DVR

Face enrollment · Attendance · Expression monitor · Safety watch

Version	Code	Branch	Date
1.0.0 (development)	amit5042k/class-gaurd	claude/camera-management-desktop-app-bR07t	October 2026

1 · AT A GLANCE

What is ClassGuard?

ClassGuard is a Windows / macOS / Linux desktop app, similar to Hikvision iVMS-4200, that brings cameras from several brands into a single screen and adds AI features on top of the video.

Screens in the app	Camera brands supported	AI features	Overall completion (estimate)
11	6	4	~55%

The main features

Live View
Watch up to 16 cameras at once in a grid (1×1 to 4×4). Streams reconnect automatically when a camera drops.

WORKING

Cameras
Add, edit or delete cameras. The app builds the correct stream address for each brand. Includes a connection test.

WORKING

NVR / DVR
Register recorders by IP and channel count. Channels are not yet turned into cameras automatically.

PARTIAL

Face Enrollment
Add people (name, ID, department) and save their face from a webcam or a photo.

WORKING

Attendance
Daily check-in / check-out report with CSV export. Automatic marking from live video is not connected yet.

PARTIAL

Expression Monitor
Detects 7 emotions (happy, sad, angry, fear, surprise, disgust, neutral) on a chosen camera.

PARTIAL

Safety Watch
Detects people in view. Detecting helmets, masks and vests requires a custom YOLO model.

PARTIAL

Alerts
An alert list with severity levels and an acknowledge button. Nothing creates alerts yet.

NOT CONNECTED

Recordings
Lists saved video files. The recording engine exists, but nothing starts it.

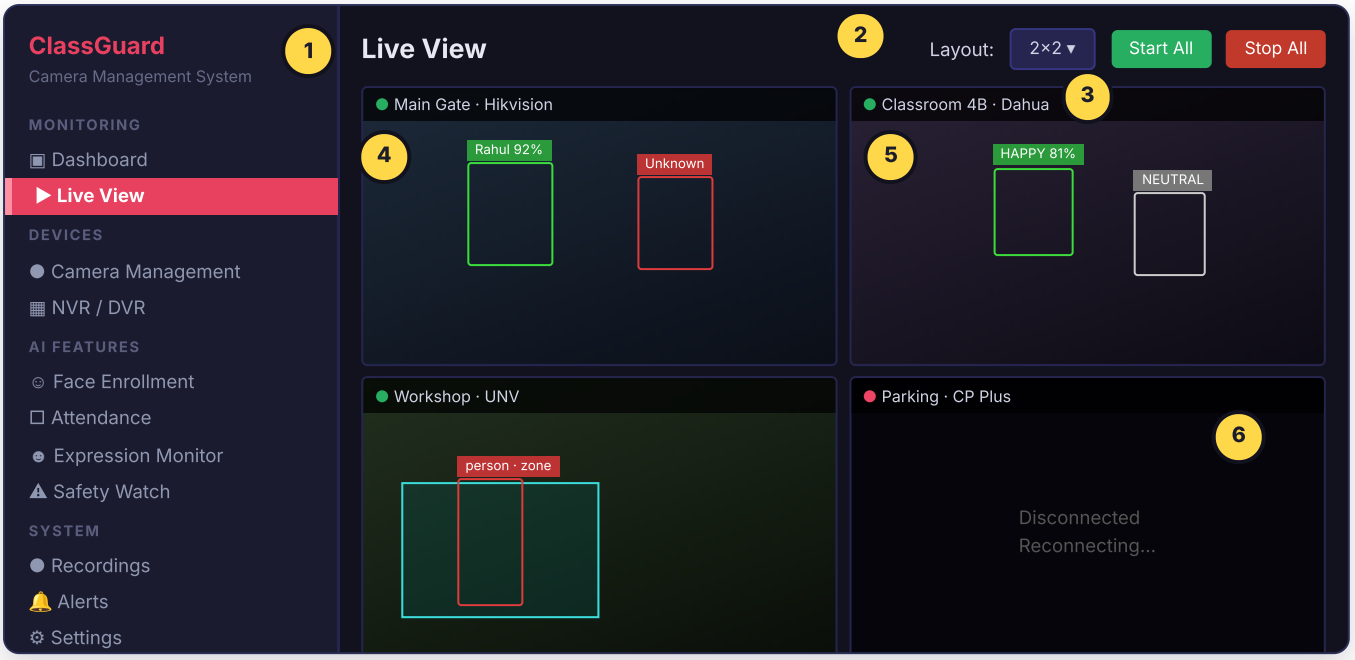
NOT CONNECTED

How to read the labels. WORKING means you can use it today. PARTIAL means the screen works but an important piece is missing. NOT CONNECTED means the parts exist but are not joined up yet.

2 · APP TOUR

Finding your way around

Every screen uses the same layout: a menu on the left and the selected screen on the right. The illustration below shows the Live View screen.



1. **Menu.** Switch between the 11 screens. The highlighted item is the current screen.
2. **Layout picker.** Choose how many cameras to show: 1×1, 2×2, 3×3, 4×4, 2×3 or 3×4.
3. **Start All / Stop All.** Connects to every active camera, or disconnects from all of them.
4. **Camera tile.** A green dot means connected. Double-click a tile, or right-click it for *Take Snapshot*.
5. **AI overlays.** Shown on the Expression and Safety screens: green box = known face or person, red box = unknown face or violation.
6. **Lost signal.** A red dot means the camera dropped. The app retries automatically, waiting a little longer each time (up to 30 s).

All 11 screens

Dashboard
Totals at a glance: cameras, streams, people, alerts, attendance, violations.

Live View
Watch cameras in a grid; take snapshots.

Camera Management
Add, edit, delete, test cameras; ONVIF scan.

NVR / DVR
Register network video recorders.

Face Enrollment
Create people and save their faces.

Attendance
Daily check-in/out list; CSV export.

Expression Monitor
Live emotion labels and chart for one camera.

Safety Watch
People / PPE / zone detection; load YOLO model.

Alerts
Review and acknowledge alerts.

Recordings
List, open and delete video files.

Settings
Recognition threshold, reconnect delay, FPS.

3 · HOW IT WORKS

Inside the app

The code is split into four layers. Each layer only talks to the layer next to it, so a single part can be changed without breaking the others.

Screens what you see	`main_window` `dashboard` `live_view` `camera_management` `nvr_management` `face_enrollment` `attendance` `expression_monitor` `safety_watch` `alerts` `recordings` `settings`
Engines the logic	`stream_manager` `camera_manager` `face_engine` `attendance_engine` `expression_engine` `safety_engine` `recording`
Brand adapters talk to cameras	`onvif_handler` `hikvision (ISAPI)` `dahua (CGI)` `generic_rtsp`
Storage remembers data	`SQLite database` `faces/` `snapshots/` `recordings/`

Path of one video frame

Each camera runs in its own background thread, so one slow camera doesn't freeze the others.

Camera
sends RTSP video

→

Stream thread
decodes each frame, reconnects if dropped

→

Live View grid
shows the picture

→

Expression & Safety
analyse the latest frame 2–3× per second

Stream thread
same frame

→

Attendance engine
should recognise faces here — **not connected yet**

→

Database
check-in / check-out rows

→

Attendance screen
shows report, CSV export

> **Red dashed boxes** mark the gap: the attendance screen can display records and the engine can write them, but no live video is sent to the engine yet. Sections 7 and 8 explain the fix.

Technology, data & troubleshooting

Reference details for the person installing or supporting the app.

Technology used

PyQt6
The desktop window, buttons and tables.

OpenCV
Reads the RTSP video, draws boxes, detects people.

SQLite + SQLAlchemy
A local database file; no server needed.

face_recognition
Turns a face into 128 numbers to match people. Optional.

DeepFace
Reads emotions from faces. Optional.

onvif-zeep / WS-Discovery
Finds ONVIF cameras on the network.

Where your data is kept

Everything stays on the local computer, in a hidden folder in your home directory: `~/.classguard/`

Path	Contents
`classguard.db`	Cameras, NVRs, people, face data, attendance, alerts
`faces/person_<id>/`	Photo saved when each face was enrolled
`snapshots/`	Snapshots from the camera grid and attendance photos
`recordings/`	MP4 video files (once recording is connected)

Troubleshooting a camera that won't connect

What you see	What to check
"Cannot open stream"	The IP address is reachable (try `ping`), port 554 is open, and RTSP is enabled on the camera.
Connects, then drops repeatedly	Weak Wi-Fi or a busy network. Try the camera's sub-stream (lower resolution).
Password contains `@` or `:`	These characters break the address. Change the password, or URL-encode it in a custom RTSP URL.

Install, then add your first camera

Install (about 10 minutes)

1. Install **Python 3.9 or newer**.
2. Download the code:
   ```
   git clone https://github.com/amit5042k/class-gaurd
   cd class-gaurd
   git checkout claude/camera-management-desktop-app-bR07t
   ```
3. Install the basic parts:
   ```
   pip install -r requirements.txt
   ```
4. *Optional:* install the full AI stack for face matching and emotions:
   ```
   pip install cmake dlib face-recognition deepface
   ```
5. Start the app:
   `python main.py` (or double-click `run.bat` on Windows)

> **No AI packages?** The app still runs. Without `face_recognition` it can only find faces, not say who they are. Without `DeepFace` every face is shown as "neutral".

Add a camera

1. Open **Camera Management** and click **+ Add Camera**.
2. Choose the **Brand**, then enter the camera's **IP address**, **username** and **password**.
3. Set the **Channel** (for a standalone camera, use 1; for an NVR, use the channel number).
4. Leave **Custom RTSP URL** empty and the app builds it for you.
5. Click **Test Connection**, then **OK**.
6. Go to **Live View** and click **Start All**.

> **Wired or Wi-Fi?** Both work the same way. The app just needs to reach the camera's IP address. The "Connection" field is only a label for your records.

Stream addresses the app builds for each brand

Brand	RTSP address template (default port 554)
Hikvision	`rtsp://user:pass@IP:554/Streaming/Channels/101`
Dahua	`rtsp://user:pass@IP:554/cam/realmonitor?channel=1&subtype=0`
UNV	`rtsp://user:pass@IP:554/media/video1`
CP Plus	`rtsp://user:pass@IP:554/h264Preview_01_main`
ONVIF	`rtsp://user:pass@IP:554/stream1` (or use ONVIF Discover)
Other	Paste the camera's own RTSP address into *Custom RTSP URL*.

> **Important:** UNV and CP Plus addresses vary between models and firmware versions. If *Test Connection* fails, look up the correct RTSP path for your model and paste it into *Custom RTSP URL*.

5 · AI FEATURES

Faces, attendance, expressions, safety

Face enrollment → Attendance

1 · Add person
name, employee ID, department

→

2 · Capture face
webcam or photo file

→

3 · Store "face print"
128 numbers saved in the database

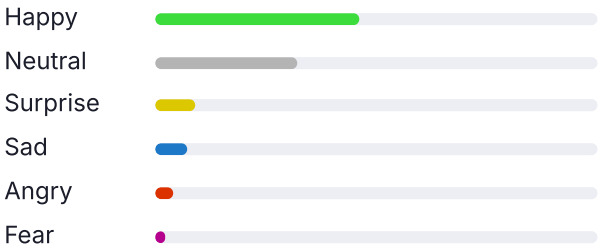
→

4 · Camera sees them
match → check-in (not connected yet)

- **Matching:** two face prints are compared by "distance". At or below **0.50** (changeable in Settings) counts as a match. Lower = stricter.
- **Attendance rule (as designed):** first sighting of the day = check-in. A later sighting after at least 60 s = check-out. A snapshot is saved with each event.
- **Tip:** enroll using a clear, front-facing, well-lit photo. Only one photo per person is stored at present.

Expression Monitor

1. In **Live View**, click **Start All** (this screen uses the same live streams).
2. Open **Expression Monitor**, choose a camera, click **Start Monitoring**.
3. Each face is labelled with its emotion. The bars show how often each emotion has appeared.



Example chart

Safety Watch

1. Start the streams in **Live View**.
2. Open **Safety Watch**, choose a camera, click **Start Safety Watch**.
3. By default it detects **people**.
4. To detect missing PPE, click **Load YOLO Weights** and pick a model trained with classes such as `no_helmet`, `no_mask`, `no_vest`.

> A PPE model is **not included**. You need to supply one in Darknet format (`.weights` + `.cfg` + `.names`). Restricted zones are supported by the engine, but there is no screen yet for drawing them.

Box colours

Colour	Meaning
GREEN	Known person (face matched), or a person with no violation
RED	Unknown face, PPE violation, or someone inside a restricted zone
CYAN	Outline of a restricted zone

> **Privacy:** face prints count as biometric data. Before using face recognition or emotion analysis on students or staff, tell the people being recorded, get any consent your local laws require, and limit who can access the computer.

6 · PROJECT STATUS

What's done and what isn't

Completion estimates are based on a code review. Nothing has been tested with real cameras yet, so expect some fixes once real hardware is connected.

Module	Status	Progress	Done ✓	Still to do
Camera management	WORKING	80%	Add/edit/delete, 6 brands, auto RTSP URL, AI toggles per camera, connection test	Fix test popup (crash risk); rebuild URL when IP changes; enable/disable camera
Live view	WORKING	75%	Grids up to 4×4, auto-reconnect, snapshots, status dots	Fullscreen, start/stop single camera, PTZ buttons, camera groups
Face enrollment	WORKING	85%	People add/edit/delete, webcam or file enroll	Multiple photos per person, enroll from IP camera, photo preview
ONVIF discovery	PARTIAL	50%	Scans network, lists devices	"Add" button on results; safe UI update from scan thread
NVR / DVR	PARTIAL	40%	Add/delete NVR records	Edit; auto-create a camera per channel; NVR playback
Attendance	PARTIAL	40%	Engine, daily report, date picker, CSV export	**Send live video to the engine**; correct check-out; shifts, late marks, monthly reports
Expression monitor	PARTIAL	65%	Live labels, emotion bars, on-screen log	Run analysis off the UI thread; save logs to database; alerts
Safety watch	PARTIAL	50%	Person detection, YOLO loader, zone logic, log	Zone drawing screen; save violations; bundled PPE model; ONNX support
Settings	PARTIAL	45%	Threshold, reconnect delay, FPS	Save settings to disk (they reset on restart)
Alerts	NOT CONN.	30%	Screen, filters, acknowledge	Create alerts from AI events; popup / sound / email
Recording & playback	NOT CONN.	25%	Recorder engine, file list	Record button and auto-record; video player; timeline; disk cleanup
PTZ control	NOT CONN.	20%	Hikvision & Dahua API code	On-screen arrow and zoom controls
Security & users	TO DO	0%	—	App login, user roles, encrypted camera passwords
Installer & tests	TO DO	5%	setup.py, run scripts	.exe installer (PyInstaller), automated tests, real-camera testing

7 · KNOWN ISSUES

Bugs: fixed and still open

Already fixed

Problem	What was wrong → fix
Expression & Safety screens showed nothing	Video frames were never sent to these screens → now forwarded from Live View.
New cameras didn't appear in Live View	Camera Management didn't notify the main window → now sends a "cameras changed" signal.
Strange "ů4 Alerts" menu label	A wrong character code → now shows the 🔔 bell.
A line of code that did nothing	Removed and replaced with the correct connections.

Still open — ranked by importance

#	Severity	Problem	Effect and suggested fix
1	HIGH	Attendance never records automatically	`AttendanceEngine.process_frame()` is never called. Call it, in a worker thread, for frames from cameras with "Attendance Tracking" turned on.
2	HIGH	Possible crash from popups in background threads	*Test Connection* and *ONVIF Discover* update the screen from a background thread, which Qt does not allow. Send results back to the UI with a signal instead.
3	HIGH	Screen freezes during AI analysis	Expression and Safety analysis runs on the UI thread; DeepFace can take hundreds of milliseconds per frame. Move it into a `QThread`.
4	MEDIUM	Possible garbled frames or crashes	The video image is built on top of a temporary memory buffer that can be freed too early. Add `.copy()` when creating the `QImage`.
5	MEDIUM	Editing a camera's IP keeps the old address	The edit form re-saves the previously generated RTSP URL. Rebuild the URL when host, port, user, password or channel changes.
6	MEDIUM	Check-out time is wrong	Check-out is set at the first sighting 60 s after check-in and never updated. It should be updated to the *last* time the person is seen.
7	MEDIUM	Per-camera AI switches are ignored	The Face / Attendance / Expression / Safety / Recording checkboxes are saved but nothing reads them.
8	MEDIUM	No alerts, violations or emotion logs saved	Database tables exist, but no code writes to them, so the Dashboard counters stay at 0.
9	LOW	YOLO file picker accepts `.pt`	OpenCV can't load PyTorch `.pt` files. Accept `.onnx` instead, and show a clear error.
10	LOW	Dahua zoom in/out reversed	Swap `ZoomWide` and `ZoomTele`.
11	LOW	Fullscreen menu item does nothing	The signal is sent but nothing listens for it. Connect it to a fullscreen viewer.
12	LOW	Passwords stored as plain text	Camera passwords are readable in the database file. Encrypt them, e.g. with the OS keyring.

8 · WHAT NEEDS TO BE DONE

Roadmap to a finished product

Suggested order of work. Each phase leaves the app in a usable state. Time estimates assume one developer.

Phase 1
Stability
~1 week

- Fix open issues #2, #3, #4 and #5 (thread safety, UI freezes, frame memory, stale URLs).
- Test with at least one real camera from each brand, plus one NVR, over both wired and Wi-Fi.
- Correct the UNV and CP Plus RTSP paths using real devices; add a sub-stream option.

Phase 2
Core AI working
~2 weeks

- Add one central AI worker that receives frames and applies each camera's AI switches (#7).
- Connect attendance to live video (#1) and fix check-out (#6). Add shifts, late/absent status and monthly reports.
- Save expression logs and safety violations; create alerts from them (#8). Add a popup and sound for high-severity alerts.
- Add a screen for drawing safety zones on the camera picture.

Phase 3
iVMS features
~2–3 weeks

- Recording: per-camera record button, scheduled and motion-triggered recording, automatic deletion of old files when the disk fills.
- Playback: video player with a timeline; play back recordings stored on Hikvision/Dahua NVRs.
- NVR channel import (one camera per channel), PTZ controls, fullscreen, camera groups/tree.
- ONVIF: add a discovered camera with one click; read its stream address automatically.

Phase 4
Production
~1–2 weeks

- Login with admin / operator / viewer roles; encrypted passwords (#12); audit log.
- Save settings to disk; back up and restore the database.
- Windows installer (.exe) using PyInstaller; macOS and Linux packages.
- Automated tests; GPU acceleration for more than 8 cameras with AI enabled.

Performance guide: a typical office PC can show about 16 HD streams. Each camera with face recognition or emotion analysis turned on uses significantly more CPU. Plan for roughly 4–6 AI-enabled cameras per PC without a GPU, then measure on the real hardware.

9 · REFERENCE

Test checklist & glossary

Acceptance test checklist

Run these with real hardware before calling a release finished.

- [ ] Hikvision camera streams in Live View
- [ ] Dahua camera streams in Live View
- [ ] UNV camera streams in Live View
- [ ] CP Plus camera streams in Live View
- [ ] ONVIF Discover finds a camera on the LAN
- [ ] NVR channels 1–4 each stream correctly
- [ ] Wi-Fi camera reconnects after the router restarts
- [ ] Unplugging a camera shows red, then green after reconnecting
- [ ] 16 cameras in a 4×4 grid run for 1 hour without slowdown
- [ ] Enrolled person is recognised at 1–3 m distance
- [ ] Unknown person is shown as "Unknown"
- [ ] Check-in and check-out appear in Attendance
- [ ] CSV export opens correctly in Excel
- [ ] Expression labels change with real facial expressions
- [ ] Person in a restricted zone creates an alert
- [ ] Recording file plays in VLC
- [ ] App closes cleanly with streams running
- [ ] Data is still there after restarting the app

Glossary

RTSP	The standard way IP cameras send live video. An address looks like `rtsp://…`
ONVIF	An industry standard that lets software from one brand control cameras from another.
NVR / DVR	A recorder box. NVRs record IP cameras; DVRs record older analogue cameras. Each camera is a "channel".
Channel	The camera's number on an NVR (1, 2, 3…).
Main / sub stream	Cameras usually send a high-quality main stream and a lighter sub stream.
PTZ	Pan-Tilt-Zoom: a camera that can be moved and zoomed remotely.
ISAPI / CGI	The web interfaces Hikvision and Dahua cameras provide for settings and control.
Face print	A list of 128 numbers describing a face. Used for matching instead of comparing photos directly.
Threshold	How close two face prints must be to count as the same person (default 0.50).
YOLO	A fast object detection model. A trained version can spot helmets, masks and vests.
HOG	A simple, built-in person detector used when no YOLO model is loaded.
PPE	Personal Protective Equipment: helmet, vest, mask, gloves, goggles.
Thread	A task running in the background so the window stays responsive.

Summary. The app has its foundations in place: multi-brand camera management, live viewing, and face enrollment all work. The AI screens show results on live video. The main gaps are connecting attendance, alerts and recording to the live video, fixing the threading issues, and testing on real cameras. Phases 1 and 2 of the roadmap close most of these gaps.